

Platform Name: Smriti Sathi (স্মৃতি সাথী)

Sponsoring Ministry: Ministry of Development of North Eastern Region (MDoNER)

Primary Domain: Healthcare & Assisted Elderly Dementia Care

Bhashini Services Requested: TTS (Text-to-Speech), ASR (Speech-to-Text), NMT (Machine Translation)

Target Languages: Assamese (অসমীয়া), Bodo (বর), Manipuri (মৈতৈলোন্), English, Hindi

Target Hardware: Android Bedside Tablets (SM-T290) & Companion Smartphones

1. Problem Statement & Clinical Rationale

Dementia and Mild Cognitive Impairment (MCI) among the aging population in the **North Eastern Region (NER)** of India present unique challenges: linguistic fragmentation, rural geographic isolation, and severe shortages of specialized psychogeriatric personnel. Elderly citizens across Assam, Bodoland, and Manipur predominantly speak native regional languages (Assamese, Bodo, and Manipuri) and frequently face high rates of digital illiteracy.

Standard cognitive training applications depend entirely on high-literacy English text and touch-heavy interactions, rendering them completely inaccessible to rural elders. **Smriti Sathi bridges this divide by embedding the Government of India's Bhashini National Language Translation Mission directly into a clinically-validated cognitive care ecosystem.**

2. Detailed Use Case Breakdown for Bhashini AI Services

Our platform harnesses three foundational Bhashini AI service pillars (Dhruva API pipeline) to create an empathetic, voice-first companion:

A. Text-to-Speech (TTS) — Voice Guidance & Reminiscence

Bhashini TTS API

Operational Role: Verbal instruction generation for non-literate and visually impaired dementia patients.

- Cognitive Exercise Prompts:** Rather than forcing elders to read onscreen instructions, Bhashini TTS narrates game objectives gently in Assamese, Bodo, and Manipuri (e.g., "একে ধৰণৰ কাৰ্ড বিচাৰক / Find the matching pair").
- Audio Daily Orientation (Masterpiece):** Recites morning-to-night schedules to re-anchor chronological orientation and prevent delirium.
- Spoken Medication & Hydration Alarms:** Speaks high-priority reminders aloud at scheduled times in the elder's native dialect, ensuring time-critical adherence.
- Cultural Reminiscence Therapy:** Delivers regional folk proverbs and cultural narratives to evoke positive emotional recall and cognitive stimulation.

B. Automatic Speech Recognition (ASR) — Natural Elder Voice Input

Bhashini ASR API

Operational Role: Hands-free voice response capture for seniors with tremors, arthritis, or touch confusion.

- **Verbal Game Responses:** Elders can speak their memory answers (e.g., reciting market costs in Bodo or naming items in Manipuri) without needing touchscreen fine-motor control.
- **Voice-Driven Medication Logging:** Elders acknowledge medication intake verbally (e.g., "ঔষধ খাই ললো" / *I have taken the medicine*), which Bhashini ASR transcribes and logs in local IndexedDB.
- **Emergency Vocal Distress:** Detects spoken distress keywords in native languages to activate emergency caregiver beacons.

C. Neural Machine Translation (NMT) — Caregiver & ASHA Bridge

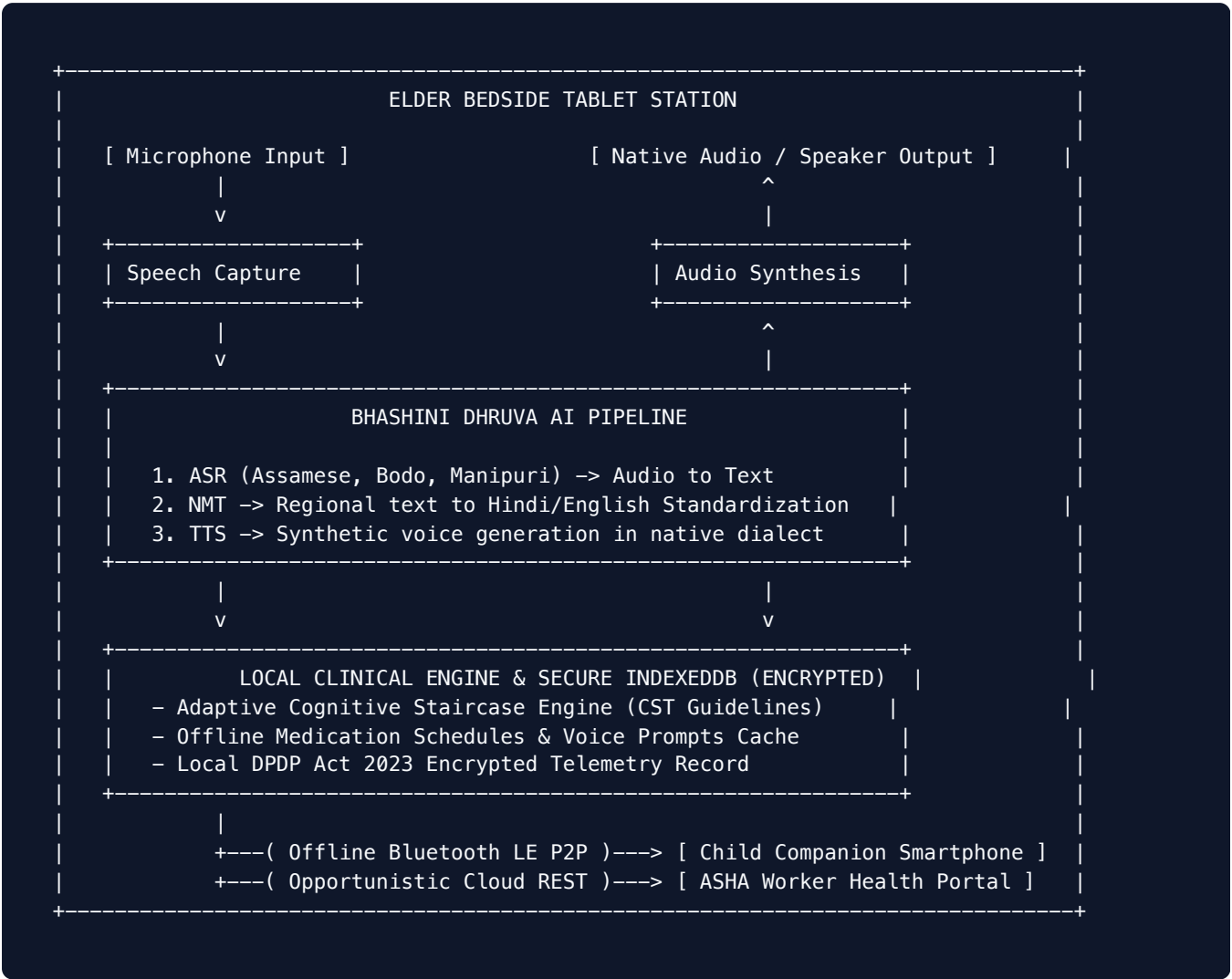
Bhashini NMT API

Operational Role: Cross-lingual harmonization between rural elders and distant family members or urban clinicians.

- Translates regional patient notes and symptom descriptions into English and Hindi for centralized medical records and ASHA coordinators.
- Enables urban family members living outside the region to configure medication regimens and reminders in English, which are translated into native Assamese/Bodo/Manipuri voice prompts on the bedside tablet.

3. System Architecture & Bhashini Pipeline

The system follows an offline-first hybrid pipeline designed specifically for low-connectivity rural healthcare environments:



4. Supported Language Matrix & Dialect Coverage

Language	Script	Bhashini Service Used	Clinical Application in Smriti Sathi
Assamese (অসমীয়া)	Bengali-Assamese	TTS, ASR, NMT	Brahmaputra Valley elder routine prompts, tea-garden market estimation games, and folk songs.
Bodo (बर')	Devanagari	TTS, ASR, NMT	Bodoland Territorial Region dialect voice assistance, daily schedule narration, and med logs.
Manipuri (মৈতৈলোন্)	Bengali / Meetei	TTS, ASR, NMT	Imphal valley cultural reminiscence proverbs, medication voice reminders, and cognitive quizzes.
English & Hindi	Latin / Devanagari	TTS, NMT	Caregiver dashboard metrics, clinical reports for doctors, and national evaluation fallback.

5. Data Privacy & Compliance (DPDP Act 2023)

Smriti Sathi treats patient voice and cognitive telemetry with the highest standard of healthcare confidentiality under the **Digital Personal Data Protection (DPDP) Act 2023**:

- **Zero Biometric Voice Retention:** Spoken audio sent to Bhashini APIs is processed transiently for recognition and immediately discarded; no patient voice samples are stored on external servers.
- **Encrypted Local Storage:** All patient identity details, cognitive assessment scores, and reminder logs are encrypted at rest using AES-GCM in local IndexedDB.
- **Revocable Caregiver Consent:** A designated family guardian maintains explicit granular consent controls to authorize or revoke data synchronization at any time.

6. Conclusion & Impact

Integrating Bhashini's AI language infrastructure transforms Smriti Sathi from a conventional screen-bound app into a culturally compassionate, voice-first medical companion for rural North Eastern dementia patients. Access to the requested Bhashini Udyat API keys will directly empower the prototype demonstration and field pilot testing for the Smart India Hackathon 2026.